Honors Track:
Competitive Programming & Problem Solving

Impartial Game Theory

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- Impartial Games
- Sprague-Grundy theorem
- Exercise problem

Game of Nim

The Game of Nim

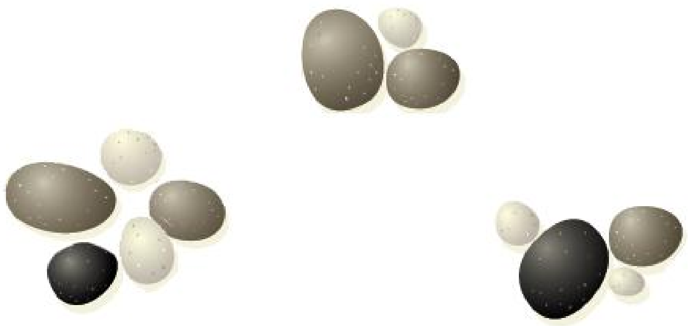
Three piles (called *nim-heaps*) of stones

Rules

- 2 players alternating turns, taking off any number of stones from a pile each turn
- Game ends when no stones are left

End condition

Player that takes the last stone(s) wins



Combinatorial Games

Rules of a Combinatorial Game

- Played by two players
- There is a (usually finite) set of possible positions of the game
- Rules of the game specify for both players and each position which moves are allowed
- Players alternate moving
- Game ends when no more moves are possible for the players whose turn it is:
  - Under **normal play rule** the last player to move *wins*
  - Under **misère play rule** the last player to move *loses*
    - *Has no strategy, very difficult to analyze*
- The game always ends in a *finite* number of moves

Impartial Game

Impartial Game

A combinatorial game, where both players have the **same set of moves**.

- So the allowable moves in an impartial game only depends on position, not which of the two players is moving.

All other combinatorial games are **partisan** ☾ next talk

Impartial Game?

Are the following games impartial games?

Tic Tac Toe

No draws in a finite number
of moves are allowed

Impartial Game?

Are the following games impartial games?

Jenga

Yes!

Impartial Game?

Are the following games impartial games?

Rock Paper Scissors

Simultaneous moves are not allowed

Impartial Game

Impartial Game

A combinatorial game, where both players have the **same set of moves**.

So the allowable moves in an impartial game only depends on position, not which of the two players is moving.

Goal

- Come up with a winning strategy

Subtraction games

Subtraction Game

- One pile of stones
- A set of positive numbers, e.g.: $S = \{1, 2\}$

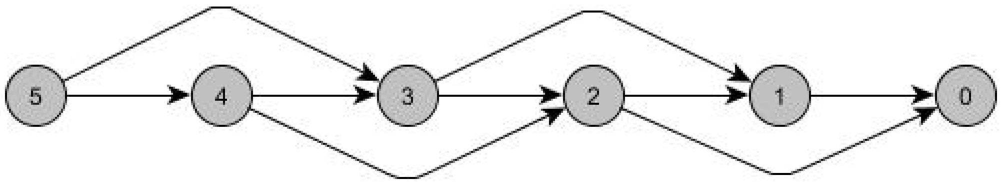
Rules

- Players alternately take n stones from the pile, where n should be a number in the set S.
- Same as Nim, the player that takes the last stone(s) from the pile wins.

Graph

Nodes are the states of the game, where in the subtraction game the state represents the amount of stones on the pile.

Edges represent the available moves.



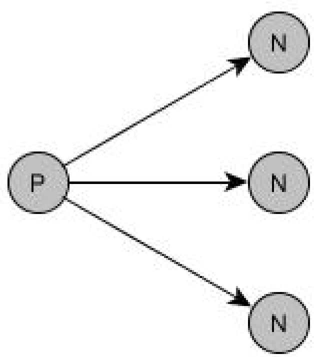
N, P positions

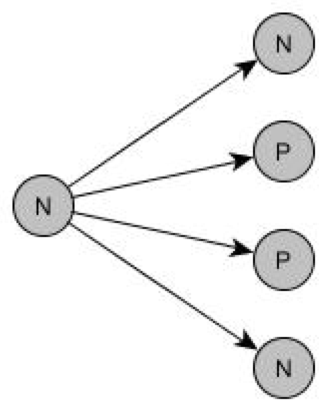
P-position

A game is in this position if it secures a win for the **P**revious player (the player that just moved).

N-position

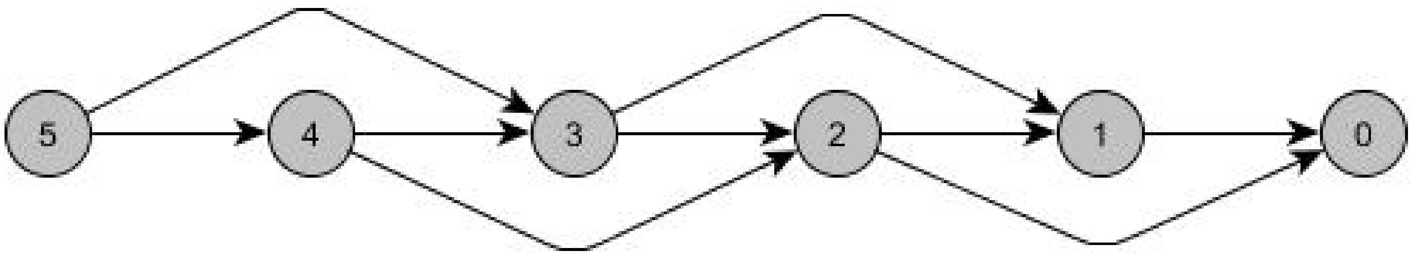
A game is in this position if it secures a win for the **N**ext player (the player that is about to make a move).





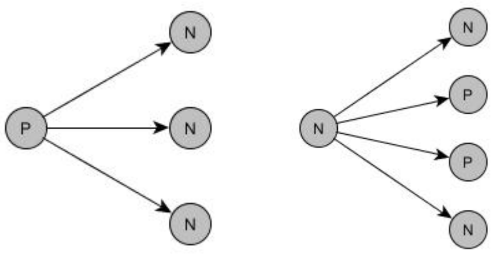
Subtraction games

Label graph with N and P positions



N N P N N P

N, P positions:



Pattern:

P N N P N N P N N …

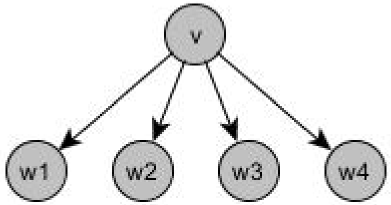
Sprague-Grundy Function

Mex (*m*inimum *ex*cludant)

- The **mex** of a set is the smallest value (≥ 0) not contained in the set
- Examples:
  - mex(∅) = 0
  - mex({1, 2, 3}) = 0
  - mex({0, 2, 4, 6, …}) = 1
  - mex({0, 1, 2, 4, 5}) = 3

Sprague-Grundy Function

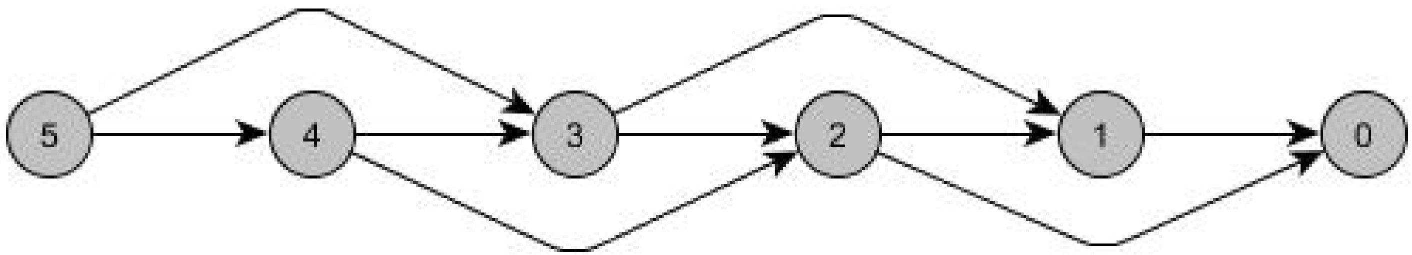
- G(v) = mex({G(w) if v ☾ w})



Sprague-Grundy Function

Sprague-Grundy Function

- G(v) = mex({G(w) if v ☾ w})



	N	N	P	N	N	P
G(v):	2	1	0	2	1	0

P-position ☾ 0

N-position ☾ **not** 0

Sum of games

Two impartial games Q_1, Q_2

Moves

You can either move in Q_1 or in Q_2

Game of Nim – single nim heap

Q_1 is one pile of stones. **Move**: take anything you like.

5	4	3	2	1	0
N	N	N	N	N	P

Sum of games

Q_2 is exactly the same game as Q_1

Now $Q := Q_1|Q_2$

Nim game with 2 heaps!

Sprague-Grundy Theorem

Nimbers

Values of the *nim-heaps*

Nimber addition ($\oplus$)

- a=5
- b=3
- Then $a \oplus b =$

$$
\begin{array}{ll}
101 & \\
011 & + \pmod 2 \\
\hline
110 & = 6
\end{array}
$$

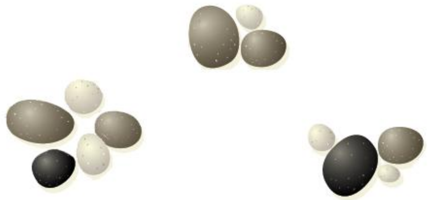
Theorem

The Sprague-Grundy function for a **sum of games** on a graph is just the **Nim sum** of the Sprague-Grundy functions of the sub-games.

Sprague-Grundy Theorem

Nim game from the start

Made of three *single-heap* Min games



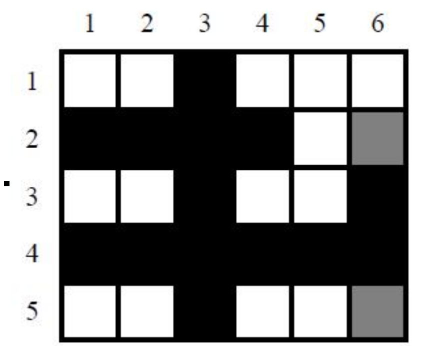
Heap 1:	101	101	001	001	001	000
Heap 2:	011	001	001	001	000	000
Heap 3:	100	100	100	+ 000	000	000
SG val:	010	000	100	000	001	000

Heap 2: -2   Heap 1: -4   Heap 3: -4   Heap 2: -1   Heap 1: -1   WIN!

Black Out – BAPC 2012 B

Black Out

- Table of 5 by 6
- Each player in turn selects one or more squares that are adjacent and in the same row or column, and colors them all black. You may paint over black squares, but at least one square needs to be white.
- The winner is the player to color the last square(s) black.
- Your AI may always start the game.



Strategy

Get to a state where you can mirror your opponent's moves (**P-position**), and the state of the table is symmetric.

You will then always be able to make a move.

How?

Start by filling squares in the 3rd row such that your filled area is centered in the middle of the row.

Exercise Problems

Impartial Games

- BAPC11G – Doubloon Game
- EAPC14E – Pawns
- EAPC11J – Shuriken Game

Further reading

- *Winning Ways for your Mathematical Plays*
  by E. Berlekamp, J.H. Conway and R. Guy (1982)